

HARSHITA BOGINENI

+91 9666409025 harshita.bogineni@gmail.com Github LinkedIn

Education

Vellore Institute of Technology

Integrated M.Tech in Computer Science (Data Science), CGPA: 9.44

2023 – Present

Vellore, India

Skills

Programming: Python, Java, C/C++

Machine Learning: Scikit-learn, TensorFlow, Explainable AI (SHAP, LIME), Soft Computing, Time-Series Analysis

Data & Statistics: Pandas, NumPy, Data Visualization, Probability, Regression, Hypothesis Testing

Core: Data Structures & Algorithms, SQL, Operating Systems, Computer Networks

Web/Tools: HTML, CSS, JavaScript, React, FastAPI, Firebase, Git, Docker

Cloud/DevOps: Kubernetes, Prometheus, Grafana

Projects

Multi-Agent AI for Appointment Optimization & Workload Balancing — [GitHub Link](#)

Python, FastAPI, LLM APIs, Machine Learning, Redis, PostgreSQL

- Designed a multi-agent healthcare scheduling system to reduce no-shows and optimize appointment allocation.
- Built predictive models to identify high-risk no-show cases and trigger proactive interventions.
- Developed agent-based workflows for risk prediction, communication, dynamic rescheduling, and waitlist optimization.
- Integrated LLM-based personalized reminders and fallback communication strategies.
- Achieved 27% reduction in no-shows and improved scheduling efficiency.

XAI-Driven Drug Intelligence: Predicting Efficacy & Toxicity with Transparent Models — [GitHub Link](#)

Python, Scikit-learn, SHAP, LIME, RDKit, Pandas, NumPy, Streamlit

- Built an end-to-end ML pipeline to predict drug efficacy and toxicity using ChEMBL and Tox21 datasets.
- Engineered interpretable molecular descriptors and handled class imbalance and feature scaling.
- Implemented multi-output models (Random Forest, Decision Trees) achieving 90% predictive performance.
- Applied SHAP and LIME for explainability, identifying key features influencing predictions.
- Developed an interactive dashboard for candidate scoring and interpretability.

SmartBMS-AI: Adaptive Battery Monitoring & Failure Prediction System — [GitHub Link](#)

Python, TensorFlow, Time-Series Analysis, Fuzzy Logic, Genetic Algorithms

- Built a physics-informed LSTM model to predict battery health and remaining useful life.
- Designed a fuzzy logic controller for safe and adaptive power regulation under uncertainty.
- Implemented a genetic algorithm optimizer to balance energy cost and system safety.
- Enabled real-time monitoring, adaptive control, and early fault detection.
- Validated performance under simulated stress scenarios (heat, load spikes).

UAMAS: Robust Multi-Agent Scaling under Uncertain Workloads — [GitHub Link](#)

Python, FastAPI, Docker, Kubernetes, Prometheus, Grafana, Machine Learning

- Developed a microservice-based framework for uncertainty-aware resource allocation and auto-scaling.
- Combined workload prediction, uncertainty modeling, and risk scoring to guide scaling decisions.
- Designed multi-agent heuristics for performance, cost, and stability optimization.
- Reduced unnecessary scaling actions while maintaining low latency (150 ms).
- Outperformed reactive and threshold-based baselines in efficiency and scaling stability.

Achievements

- Smart India Hackathon 2025 (College Level) – Developed AI-based dropout prediction system.
- Participated in Caterpillar Code-a-Thon and worked on a real-world problem-solving project.
- Built multiple academic projects in AI/ML, including healthcare assistant, battery prediction, and resource allocation systems.

Relevant Coursework

- | | | |
|--------------------|-------------------------------|---------------------|
| • Machine Learning | • Foundations of Data Science | • Computer Networks |
| • Cloud Computing | • Operating Systems | • Soft Computing |

Leadership & Activities

- Program Representative (2023–2024): Coordinated between students and faculty, improving communication and resolving academic issues.

Languages

- English, Telugu, Hindi, Tamil